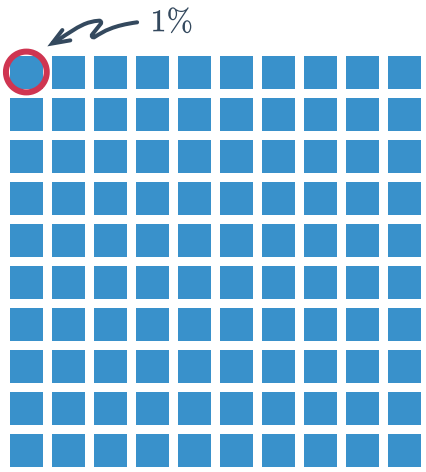
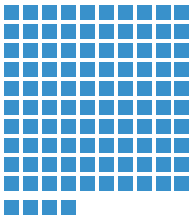
Addition and subtraction in percentage change

1 Here is a grid of 100 blue squares. Each small square represents 1% of the total:

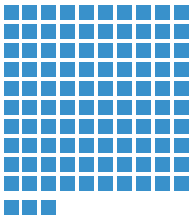


a Choose the image that represents 3% more than the original grid:

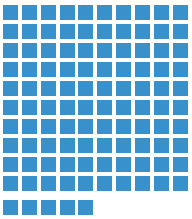
A



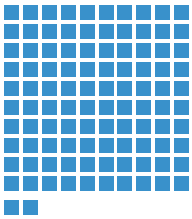
B



C



D

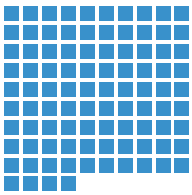


b Describe what was done to the original grid to increase it by 3%.

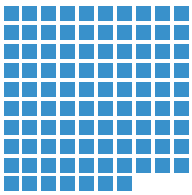
c Write an expression to represent the 3% increase in squares.

d Choose the image that represents 6% less than the original grid:

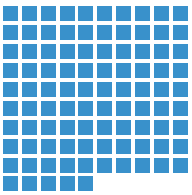
A



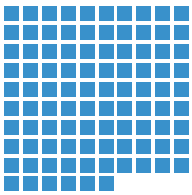
B



C



D



e Describe what was done to the original grid to decrease it by 6%.

f Write an expression to represent the 6% decrease in squares.

2 Bob wants to decrease 110 by 60%.

a Calculate 60% of 110.

b Should Bob add or subtract the amount from part (a) from 110? Explain your answer.

c Hence, decrease 110 by 60%.

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3 Lucy wants to increase 70 by 60%.

- **a** Calculate 60% of 70.
- **b** Should Lucy add or subtract the amount from part (a) from 70? Explain your answer.
- **c** Hence, increase 70 by 60%.

Percentage increase

4 Increase:

- **a** 80 by 5%.
- **b** 120 by 10%.
- **c** 72 by 40%.
- **d** 320 by 25%.
- **e** $\$140$ by 9%.
- **f** $\$30\,000$ by 20%.
- **g** 80 km by 15%.
- **h** 400 mL by 35%.

5 Abed wants to increase 1500 by 30%.

- **a** Calculate 30% of 1500 and add the answer to 1500.
- **b** Calculate 130% of 1500.
- **c** Are your answers to (a) and (b) the same? Explain why this is.

6 Tracy wants to increase 80 by 30%.

- **a** What single percentage should she multiply 80 by?
- **b** Explain how you got your percentage in part (a).

7 What single percentage should a quantity be multiplied by in order to increase the quantity by:

- **a** 17%
- **b** 5%
- **c** 40%
- **d** 9.4%
- **e** 33%
- **f** 127%
- **g** 38%
- **h** 212%

8 Evaluate the following using a single percentage:

- **a** Increase 80 by 60%
- **b** Increase 400 by 45%
- **c** Increase 900 by 8%
- **d** Increase 80 by 40%
- **e** Increase 1200 by 45%
- **f** Increase 1200 by 8%

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9 A bag of rice weighs $120\ \text{kg}$. The company who makes these bags decides to increase their weight by 45%. Find the new weight of the bag.

Percentage decrease

10 Decrease:

a 70 by 8%.

b 260 by 10%.

c 96 by 60%.

d 250 by 20%.

e $\$480$ by 18%.

f $\$55\,000$ by 45%.

g $200\ \text{km}$ by 22%.

h $650\ \text{mL}$ by 30%.

11 Derek wants to decrease 120 by 70%.

a What single percentage should he multiply 120 by?

b Explain how you got your percentage in part (a).

12 What single percentage should a quantity be multiplied by in order to decrease the quantity by:

a 17% **b** 5% **c** 40% **d** 9%

e 32% **f** 52% **g** 38% **h** 65%

13 Evaluate the following using a single percentage:

a Decrease 80 by 60%

b Decrease 400 by 45%

c Decrease 900 by 8%

d Decrease 80 by 40%

e Decrease 1200 by 45%

f Decrease 1200 by 8%

14 A discus thrower has a personal best of $60\ \text{m}$. In their most recent competition they fall short of their personal best by 30%. Find the distance of their throw in this competition.

Percentage change

15 A percentage increase was applied to the number 60 in order to get 108.

a Find how much was added to 60 to get 108.

b Express the increase as a percentage of 60.

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16 A percentage increase was applied to the number 180 in order to get 207.

a Find how much was added to 180 to get 207.

b Express the increase as a percentage of 180.

17 A percentage decrease was applied to the number 90 in order to get 27.

a Find how much was subtracted from 90 to get 27.

b Express the decrease as a percentage of 90.

18 A percentage decrease was applied to the number 180 in order to get 117.

a Find how much was subtracted from 180 to get 117.

b Express the decrease as a percentage of 180.

19 In training for her next marathon, Sally increased her practise route from 7000 m to 7910 m. What percentage has Sally increased the distance of her practice route by?

20 Luke's record for juggling a football was 70 keep ups. After buying some new football boots, this increased to 112. Express this increase as a percentage of the original record.

21 Sharon's daily pay was $\$150$. When she began working shorter hours in order to study, her pay decreased to $\$90$ per day. Express this decrease as a percentage of her original pay.

22 Jimmy had a bowling high score of 60 points. After practising for a week, this increased to 93. Express this increase as a percentage of his original high score.

23 Sophia is able to write 220 words in a minute. After sustaining a thumb injury, this decreased to 143 words a minute. Express this decrease as a percentage of her original rate.

24 During the editing phase when making a movie, the director ordered that the movie be cut down from 200 minutes to 124 minutes. What percentage did the movie's duration decrease by from its original length?

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